

Analysis of Vehicular Fuel Efficiency: Statistical Modeling and Diagnostic Insights

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Abstract

This study examines the Auto MPG dataset through both simple and multiple linear regression frameworks. By identifying key predictors of fuel efficiency and evaluating the impact of variable interactions and non-linear transformations, we demonstrate that a logarithmic approach to vehicle weight significantly enhances model interpretability and predictive accuracy.

1 Theoretical Background

1.1 Evaluation of Independent Variable Significance

In the context of multiple regression, p -values associated with coefficients test the null hypothesis (H_0) that a predictor has no marginal effect on the response variable when all other covariates are held constant.

- **Significant Predictors:** TV and Radio advertising show $p < 0.0001$, indicating they provide unique explanatory power to sales.
- **Redundant Predictors:** Newspaper advertising ($p = 0.8599$) indicates a lack of significant linear association once the effects of TV and Radio are accounted for.

1.2 Hat Matrix Derivation for Intercept-free Models

For a model without an intercept, we define the leverage of observations through the predicted values $\hat{y}_i = x_i \hat{\beta}$. The OLS estimator is:

$$\hat{\beta} = \frac{\sum_{i'=1}^n x_{i'} y_{i'}}{\sum_{i'=1}^n x_{i'}^2}$$

By substitution, we find the linear combination of $y_{i'}$ that forms the prediction:

$$\hat{y}_i = x_i \left(\frac{\sum_{i'=1}^n x_{i'} y_{i'}}{\sum_{j=1}^n x_j^2} \right) = \sum_{i'=1}^n \left(\frac{x_i x_{i'}}{\sum_{j=1}^n x_j^2} \right) y_{i'}$$

Thus, the leverage or influence of the i' -th observation on the i -th prediction is determined by the weight $a_{ii'} = \frac{x_i x_{i'}}{\sum_{j=1}^n x_j^2}$.

2 Single-Predictor Modeling: Horsepower

2.1 Correlation and Predictive Inference

Using *horsepower* as the sole predictor, the model yields $\text{mpg} = 39.94 - 0.158(\text{horsepower})$. The results indicate a strong negative relationship ($R^2 = 0.6059$) where higher engine power correlates with reduced fuel efficiency. For a vehicle with 98 horsepower, the model predicts an MPG of 24.47.

Figure 1: MPG vs Horsepower Regression Fit

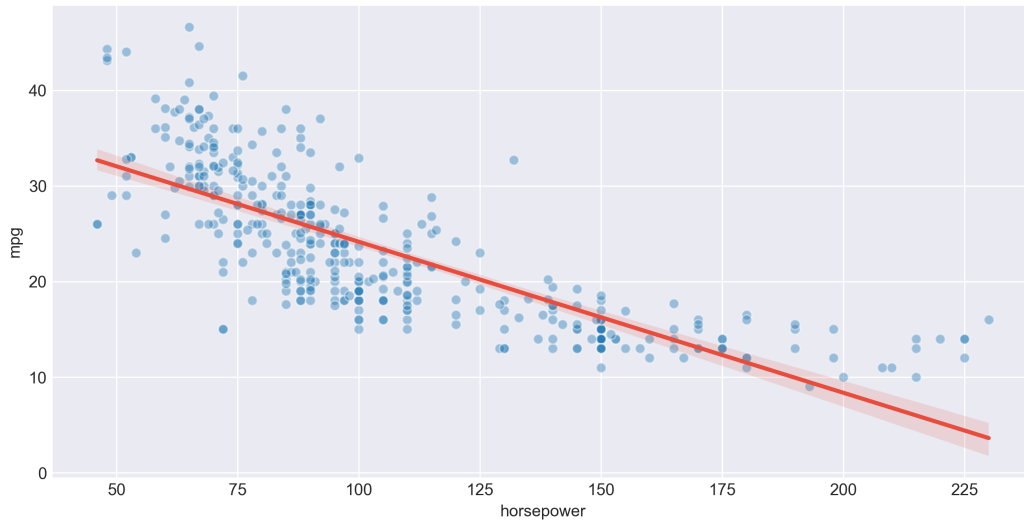


Figure 1: Regression profile of MPG against Horsepower showing a clear inverse trend.

2.2 Diagnostic Assessment

The diagnostics in Figure 2 highlight the limitations of the linear assumption. The *Residuals vs Fitted* plot exhibits a non-random, parabolic pattern, suggesting that the underlying relationship is likely non-linear (e.g., polynomial).

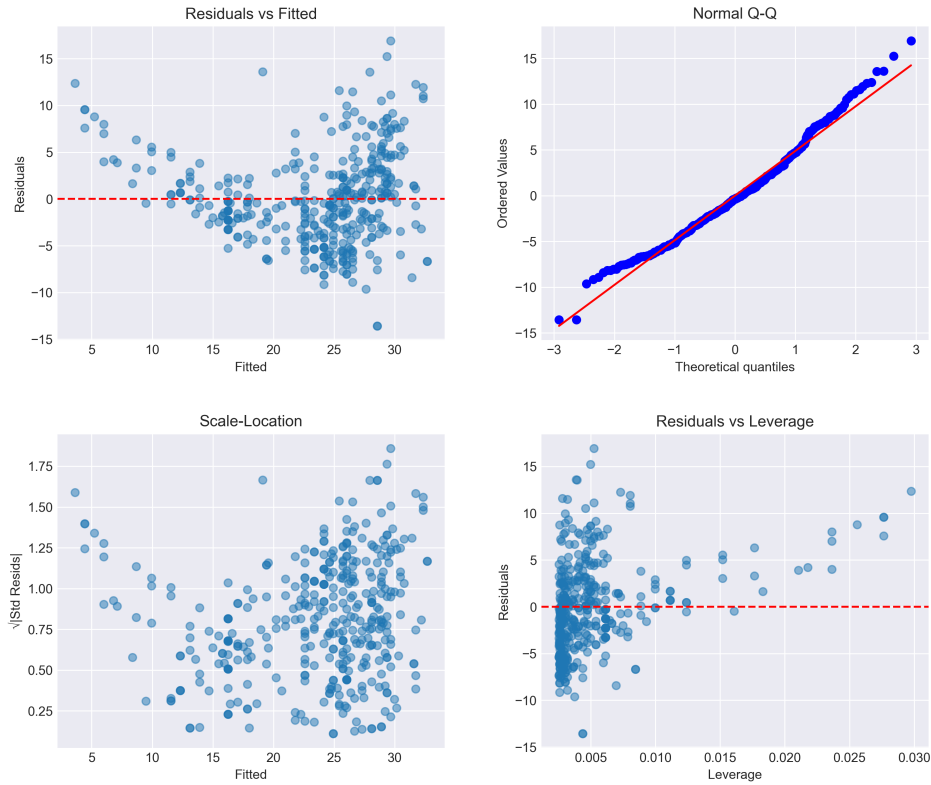


Figure 2: Diagnostic panels for SLR: Evidence of non-linearity and heavy-tailed residuals.

3 Multivariate Analysis and Feature Engineering

3.1 Exploratory Matrix and Collinearity

Figure 3 illustrates the complex inter-dependencies between predictors. Notably, *displacement*, *weight*, and *horsepower* are highly collinear, which can destabilize coefficient estimates.

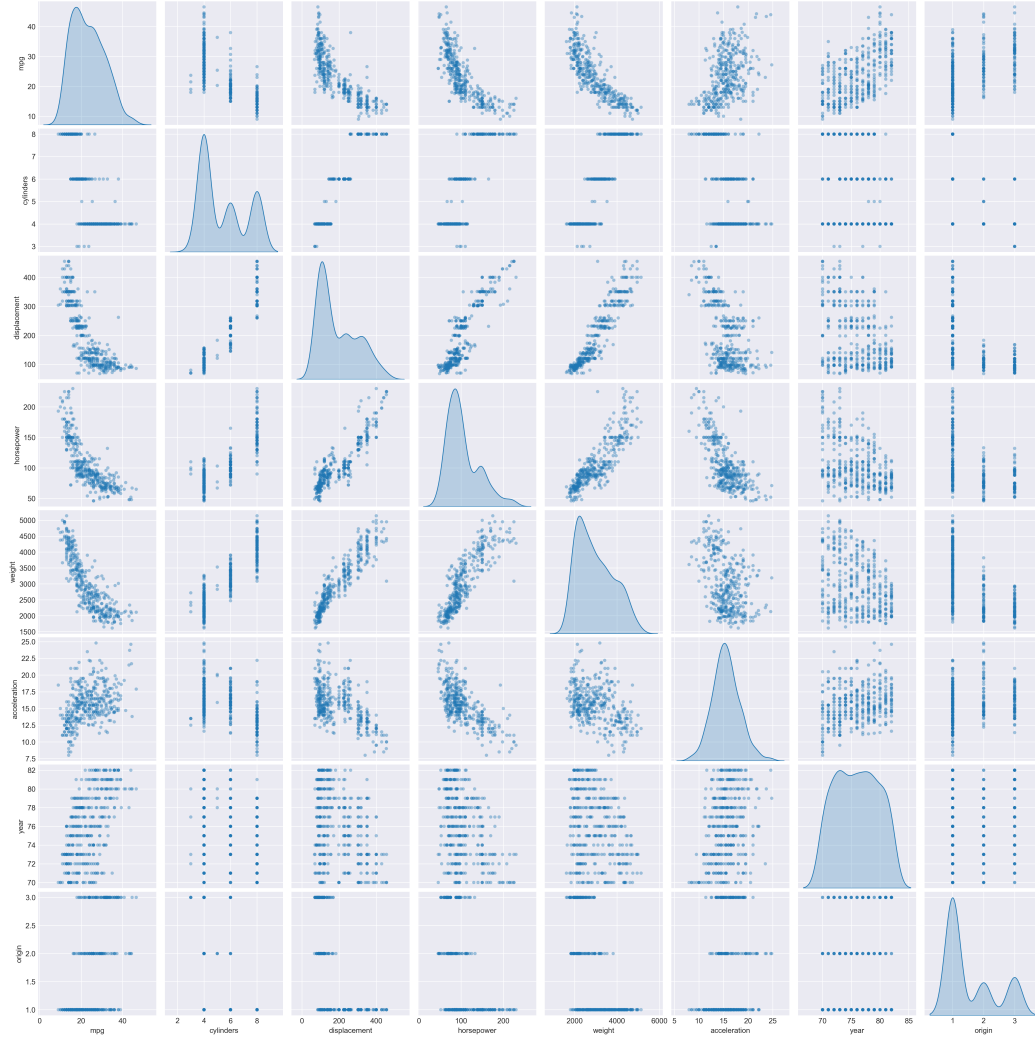


Figure 3: Scatterplot matrix detailing the multivariate distributions within the dataset.

3.2 Multiple Linear Regression (MLR) Results

The full model achieves an adjusted R^2 of 0.8182. Significant drivers include *weight*, *year*, *origin*, and *displacement*. Interestingly, the correlation heatmap (Figure 4) confirms that while many variables correlate with MPG, their marginal contributions vary significantly.

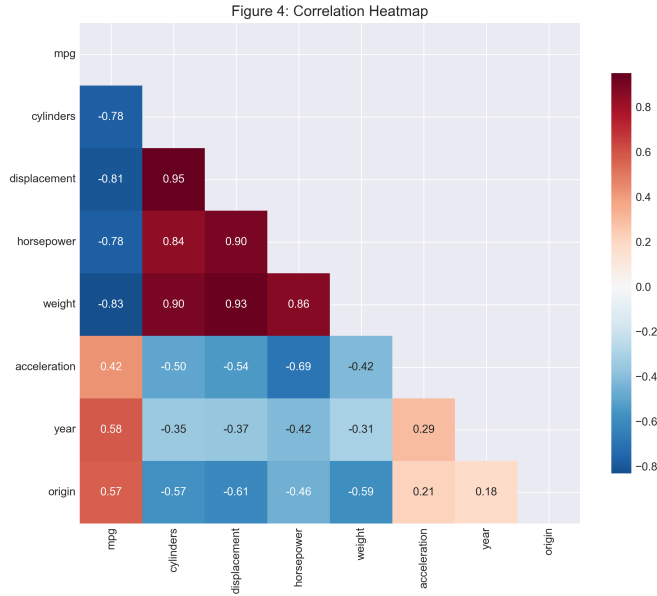


Figure 4: Correlation Heatmap: Blue tones indicate negative associations with MPG.

3.3 Diagnostics and Leverage Points

The MLR diagnostics (Figure 5) identify specific influential observations (high leverage) and confirm moderate heteroscedasticity.

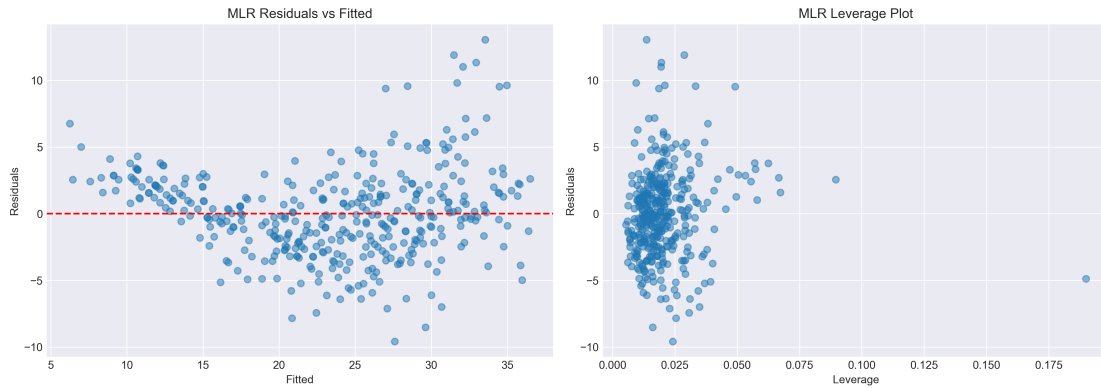


Figure 5: MLR Residual Analysis: Identification of outliers and high-leverage instances.

4 Model Optimization

4.1 Interaction Effects

Statistical testing revealed critical synergies. Specifically, the interactions between *displacement* and *weight*, as well as *horsepower* and *weight*, were highly significant ($p < 0.001$). This implies that the impact of vehicle weight on fuel economy is not constant but depends on the engine's power characteristics.

4.2 Refinement via Logarithmic Transformation

A logarithmic transformation of the *weight* variable proved the most effective enhancement, raising the R^2 to 0.8459. This suggests that MPG changes more proportionally to the percentage change in weight rather than the absolute change.

5 Final Summary

The analysis confirms that while engine power is a strong indicator of fuel consumption, a multivariate approach incorporating vehicle weight and technological evolution (model year) provides a much more robust explanation. Addressing non-linearity through log-transformations remains essential for capturing the true dynamics of automotive fuel efficiency.